

The Relationship Between Study Application Usage Frequency and Academic Performance of Second-Year BSIT Students at National University-Manila

Francis Elijah Baligod
College of Computing and
Information Technologies
National University–Manila
Manila, Philippines
baligodfc@students.national-u.edu.ph

Zyrell Grace De Guia
College of Computing and
Information Technologies
National University–Manila
Manila, Philippines
deguiazc@students.national-u.edu.ph

Mary Reched Dizon
College of Computing and
Information Technologies
National University–Manila
Manila, Philippines
dizonml@students.national-u.edu.ph

Kathleen Nea Macaspac
College of Computing and
Information Technologies
National University–Manila
Manila, Philippines
macaspacka@students.national-u.edu.ph

Abstract— This study examined the relationship between study application usage frequency and the academic performance of second-year Bachelor of Science in Information Technology students at National University–Manila. A quantitative descriptive-correlational design was used, and data were collected through a structured online questionnaire answered by 100 respondents from the Mobile and Web Applications and Multimedia Arts and Animation specializations. The questionnaire gathered data on respondents’ study application usage, commonly used application categories, perceived learning impact, and self-reported General Weighted Average (GWA) range. Descriptive statistics were used to summarize the responses, while Pearson Product-Moment Correlation was used to determine the relationship between coded study application usage frequency and coded GWA range. Results showed that most respondents used study applications for academic purposes, with collaboration tools, learning platforms, and note-taking applications as the most common categories. The respondents also generally agreed that study applications helped them complete academic tasks, improve lesson understanding, increase productivity, and support academic performance. The Pearson correlation result showed a moderate positive and statistically significant relationship between study application usage frequency and GWA range, with an r -value of 0.58 and a p -value of less than 0.001. Therefore, the null hypothesis was rejected. The findings suggest that more frequent use of study applications is associated with higher GWA ranges among the respondents. However, since convenience sampling was used and academic performance was based on self-reported GWA range, the results should be interpreted within the context of the respondents and should not be generalized to the entire second-year BSIT student population.

Keywords— academic performance, General Weighted Average, Pearson correlation, study application

I. INTRODUCTION

A. Background of the Study

Digital technologies have become a regular part of education, especially as students now rely on different online tools to support their academic work. Aside from traditional classroom activities, many students use mobile learning applications and digital study tools to access learning materials,

review lessons, organize tasks, and complete school requirements. Demir and Akpınar found that mobile learning applications may help improve students’ academic achievement, attitudes toward mobile learning, and motivation, which shows that these tools can play a meaningful role in the learning process when used properly [1].

For Bachelor of Science in Information Technology (BSIT) students, the use of study applications is especially relevant because many of their academic tasks are already connected to technology. Their coursework often involves programming, research, digital outputs, and project-based activities, which require them to use online platforms and other learning tools. In a related study, Bayar and Kurt found that mobile-supported learning improved students’ academic achievement in a science course, suggesting that mobile learning can help students understand lessons better when it is used as part of structured learning activities [2]. Although their study focused on a different group of learners, it still supports the idea that mobile learning tools can assist students in academic tasks.

However, the use of digital tools does not automatically lead to better academic performance. Their effect may depend on how often students use them, what purpose they serve, and whether they are used for academic or non-academic activities. Viloan et al. found that digital technology influences students’ perceived academic performance, showing that students recognize the role of technology in their learning experiences [3]. Meanwhile, Ang et al. studied screen time and academic performance and highlighted that technology use may have different effects depending on whether screen time is used for school-related or non-school-related purposes [4]. This suggests that the relationship between technology use and academic performance should not only consider access to digital tools, but also the way students use them.

In the Philippine setting, mobile learning has also been discussed as a tool that can support students’ access to learning, especially during situations where regular classroom interaction is limited. Bacolod explained that mobile learning helped students continue their academic activities during restricted learning conditions, although its effectiveness still depended on the learners’ situation and how the tools were implemented [5].

This shows that mobile learning and educational applications can support students, but they are not always effective in the same way for everyone.

Although previous studies have already explored mobile learning, digital technology, screen time, and perceived academic performance, there is still a need to focus more specifically on study application usage frequency. Many existing studies discuss technology use in a general sense, while fewer studies examine how often students use study applications and how this relates to actual academic performance. In addition, there is limited localized research focusing on second-year BSIT students, particularly those from specializations such as Mobile and Web Applications (MWA) and Multimedia Arts and Animation (MAA), whose academic tasks may influence their use of study applications.

At National University–Manila, second-year BSIT students face academic demands that require effective study habits and proper use of learning tools. Since study applications may help students review lessons, organize materials, practice skills, and complete academic tasks, it is important to examine how frequently these tools are used and whether their usage has a relationship with academic performance, particularly the students' General Weighted Average (GWA) range. Therefore, this study focuses on determining the relationship between study application usage frequency and the academic performance of second-year BSIT students at National University–Manila.

B. Statement of the Problem

This study aimed to determine the relationship between study application usage frequency and the academic performance of second-year BSIT students at National University–Manila. Specifically, this study sought to answer the following questions:

1. What proportion of second-year BSIT students at National University–Manila use study applications for academic purposes?
2. How frequently do second-year BSIT students at National University–Manila use study applications?
3. What categories of study applications are commonly used by second-year BSIT students at National University–Manila?
4. What is the academic performance of the respondents in terms of their General Weighted Average (GWA)?
5. What is the respondents' level of agreement regarding their study application usage behavior and the perceived impact of study applications on learning?
6. Is there a significant relationship between the frequency of study application usage and the academic performance of second-year BSIT students at National University–Manila?

C. Research Objectives

The study aimed to determine the relationship between study application usage frequency and the academic performance of second-year BSIT students at National University–Manila.

Specifically, this study aimed to:

1. Determine the proportion of second-year BSIT students who use study applications for academic purposes.
2. Identify the frequency of study application usage among the respondents.
3. Determine the categories of study applications commonly used by the respondents.
4. Assess the academic performance of the respondents in terms of their General Weighted Average (GWA).
5. Determine the respondents' level of agreement regarding their study application usage behavior and the perceived impact of study applications on learning.
6. Determine whether there is a significant relationship between study application usage frequency and academic performance.

D. Research Hypotheses

This study tested the following hypotheses:

H₀: There is no significant relationship between study application usage frequency and the academic performance of second-year BSIT students at National University–Manila.

H₁: There is a significant relationship between study application usage frequency and the academic performance of second-year BSIT students at National University–Manila.

E. Scope and Delimitations

This study determined the relationship between study application usage frequency and the academic performance of second-year Bachelor of Science in Information Technology (BSIT) students at National University–Manila. It also examined the respondents' study application usage patterns, including whether they use study applications for academic purposes, how often they use them, the average number of hours they spend using them per day, and the categories of study applications they commonly use. In addition, the study assessed the respondents' level of agreement regarding their study application usage behavior and the perceived impact of study applications on their learning. Academic performance was measured based on the respondents' self-reported General Weighted Average (GWA) range.

The study focused only on second-year BSIT students at National University–Manila, specifically those from the Mobile and Web Applications (MWA) and Multimedia Arts and Animation (MAA) specializations. A total of 100 respondents participated in the survey. Convenience sampling was used because the respondents were selected based on their

availability, accessibility, and willingness to answer the questionnaire. Data were gathered through an online questionnaire using Microsoft Forms to make the data collection process faster and more organized.

However, the study did not include students from other year levels, programs, or institutions. Since convenience sampling was used, the results may not fully represent the entire population of second-year BSIT students at National University–Manila. Therefore, the findings should be interpreted only within the context of the respondents who participated in the study and should not be generalized to all BSIT students or to the entire student population.

The study only focused on study application usage frequency, usage behavior, perceived learning impact, and academic performance in terms of GWA range. Other factors that may affect academic performance, such as motivation, study habits, learning styles, teaching strategies, socio-economic status, internet accessibility, and personal circumstances, were not treated as main variables. Since the study used a descriptive-correlational approach, it did not determine whether study application usage directly caused changes in academic performance. The data also depended on the self-reported answers of the respondents, including their GWA range, which may be affected by personal bias, recall errors, or reporting inaccuracies.

F. Significance of the Study

This study may provide a clearer understanding of how study application usage frequency relates to the academic performance of second-year BSIT students at National University–Manila. The findings may benefit BSIT students by helping them become more aware of how they use study applications for academic tasks such as reviewing lessons, completing assignments, organizing study materials, and practicing technical skills. The results may also help them reflect on whether these applications support their productivity, learning efficiency, and academic performance.

The study may also benefit professors by providing insight into how students use digital study tools in their academic work. These findings may help professors recommend appropriate applications, learning resources, or study strategies that can support students' academic needs. For National University–Manila, the study may provide useful information on students' use of study applications and their perceived role in learning, which may help in improving academic support and promoting the effective use of digital learning tools.

Lastly, this study may serve as a reference for future researchers conducting related studies on study applications, mobile learning, digital learning behavior, and academic performance.

II. REVIEW OF RELATED LITERATURE

This section presents related studies and literature on study application usage, mobile learning, digital learning tools, study habits, and academic performance. These sources provide a foundation for understanding how students use digital tools to

support academic tasks and how these tools may be connected to learning outcomes. The reviewed literature also helps establish the research gap addressed by the present study, particularly the need to examine study application usage frequency and its relationship with the academic performance of second-year BSIT students at National University–Manila.

A. Mobile Learning as a Digital Learning Tool

The increasing use of mobile devices in academic settings has led researchers to examine mobile learning as a tool that supports teaching and learning. Findings from previous review-based studies suggest that mobile learning can have a positive effect on students' academic outcomes. Yang and Xiang [6] conducted a second-order analysis of existing studies on mobile learning and students' academic achievement, showing that mobile learning generally produces positive effects across different educational contexts. Similarly, Talan [7] reviewed 104 studies and concluded that mobile learning interventions have a significant positive effect on learning performance.

Garzon et al. [8] further supported this finding through a large-scale review that combined the results of 253 studies. Their findings showed that mobile learning significantly improves students' learning gains, suggesting that mobile technologies can be effective when used for academic purposes. This is relevant to the present study because it supports the idea that mobile learning tools are not only commonly used but may also lead to measurable improvements in learning outcomes.

In addition to academic effectiveness, researchers have also studied the factors that influence the successful use of mobile learning. Jacob and Issac [9] examined students' perceptions and the role of mobile learning culture in higher education. Their study showed that students generally have positive views toward mobile learning, but its adoption is affected by the learning environment and institutional support. Similarly, Alrasheedi and Capretz [10] identified several factors that contribute to successful mobile learning, including technical readiness, motivation, usability, and instructional support. These findings suggest that mobile learning is more effective when students are motivated, the technology is accessible and easy to use, and the learning process is properly supported.

B. Study Applications and Academic Performance

Research on study applications and academic performance shows that mobile-assisted learning can support students' learning outcomes when the tools are properly designed and used for academic purposes. Bazhenova et al. [11] examined the effect of mobile learning on undergraduate students' cognitive learning outcomes. Their findings showed that mobile learning can help improve knowledge retention and higher-order thinking among university students. Similarly, Sung et al. [12] reviewed previous studies on the use of mobile devices in teaching and learning and found that mobile device integration has a significant positive effect on students' learning performance.

Toh and Tasir [13] examined the use of a mobile learning application in Biology and its effect on students' cognitive load and learning performance. Their study found that students who used a mobile application designed based on learning theories experienced lower cognitive load and achieved better learning performance compared to those who did not use the application. This suggests that the effectiveness of study applications depends not only on how often students use them, but also on whether the application is designed in a way that supports understanding and reduces difficulty in learning.

Togaibayeva et al. [14] also examined mobile learning in relation to students' satisfaction, perceived usefulness, and academic performance. Their findings showed that motivation, content relevance, and self-efficacy influenced students' academic outcomes. This indicates that mobile learning does not automatically lead to better performance. Instead, study applications become more effective when students find them useful, relevant to their learning needs, and supportive of their confidence in completing academic tasks.

C. Usage Frequency and Academic Purpose of Study Applications

The frequency of study application usage is an important factor in understanding how digital tools relate to academic performance. However, previous studies show that frequency should not be viewed separately from the purpose and pattern of use. Hsieh [15] examined smartphone usage frequency and academic performance among university students and found that smartphone use may affect academic performance depending on how and why the device is used. This suggests that frequent use of mobile devices does not automatically support learning, especially when the use is not related to academic tasks. This is relevant to the present study because it focuses specifically on study application usage rather than general smartphone use.

Mushtaq [16] also examined smartphone usage patterns and academic performance among university students. Although many students used smartphones for several hours a day, the study showed that the amount of time spent using smartphones did not fully explain academic performance. Instead, the purpose of use and the activities performed on the device were more important factors. This supports the idea that frequency should be interpreted together with how students use applications for academic tasks.

Ahmed et al. [17] provided a more application-specific perspective by examining how different app categories and their usage frequency relate to students' CGPA. Their findings suggested that certain types of applications may be linked to academic performance, showing that not all forms of digital usage have the same academic value. Similarly, Tus et al. [18] examined Learning Management System usage among Filipino students during online learning. Although students reported using and appreciating LMS tools, the study did not find a significant relationship between LMS usage and academic

performance. This suggests that the frequent use of digital learning platforms does not automatically lead to better academic outcomes.

Recent studies also show that the effect of usage frequency may differ depending on the learning context. Anwar et al. [19] studied engineering and physics students who used CourseMIRROR, a mobile reflection application. The study measured app engagement based on how often students viewed summaries of their reflections. Their findings showed that app engagement was significantly related to final grades in the engineering course, while academic engagement was more strongly related to performance in the physics course. This indicates that the value of study application usage may depend on the subject, course design, and the way the application supports learning.

Sharaf-Addin et al. [20] also examined mobile application use among English language undergraduates. Although most students reported that mobile applications helped improve their English learning, the study found that frequency of use did not create a statistically significant difference in perceived learning outcomes. This suggests that regular use of study applications does not automatically lead to academic improvement. Other factors, such as motivation, app quality, learning strategy, and relevance to the course, may also influence the results.

Johannsen et al. [21] emphasized the role of user satisfaction and system quality in mobile learning. Their study found that learning effectiveness was influenced by students' satisfaction and intention to continue using the application, which were affected by perceived enjoyment, system quality, and information quality. This supports the idea that students are more likely to use study applications consistently when the tools are useful, easy to use, and enjoyable. Therefore, study application usage frequency should be understood not only as a study habit, but also because of positive user experience and effective application design.

D. Categories and Academic Functions of Study Applications

Identifying the categories of study applications used by students is important because different applications support different academic needs. Hinze et al. [22] examined mobile application use for teaching and research in higher education and identified several types of study-related applications. These included document and data storage tools, communication applications, reference and information-retrieval tools, and productivity applications for scheduling and task management. Their study also found that students often use applications for supporting academic tasks, such as organizing files and communicating with others, rather than for direct classroom instruction.

Rangel-de Lazaro and Duarte [23] also provided a framework for classifying mobile learning applications in online higher education. Their review identified content

creation, communication and collaboration, and assessment tools as major categories of mobile learning applications. These categories are relevant to the present study because they can help organize the types of study applications used by the respondents, such as note-taking applications, learning management systems, collaboration platforms, quiz applications, and productivity tools.

Gupta et al. [24] categorized mobile learning technologies according to how they support instruction. Their review emphasized the difference between synchronous tools, such as video conferencing and live chat, and asynchronous tools, such as recorded lectures, self-paced modules, discussion forums, and learning platforms. This distinction is useful in the present study because students may use study applications differently depending on whether the learning activity requires real-time participation or independent study.

In the Philippine context, Luzano et al. [25] examined how second-year college students used Symbolab and Photomath in learning calculus. Their findings showed that students viewed these applications as learning support tools because they were accessible, user-friendly, and helpful in understanding mathematical problems. However, the study also emphasized the importance of ethical use, suggesting that study applications can support learning when they are used responsibly. This is relevant to BSIT students because their academic tasks often involve problem-solving, independent learning, research, and the use of digital tools for technical subjects.

Overall, previous studies show that study applications can be classified based on their academic function, such as learning management, communication, collaboration, content creation, assessment, reference, productivity, and subject-specific learning support. These categories are important in the present study because the academic value of study applications may vary depending on the type of application used and the purpose for which students use it.

E. Study Applications, Learning Habits, and Academic Productivity

The relationship between study application usage and academic productivity is influenced by students' learning habits and strategies. Babatunde [26] examined mobile application usage and academic performance, with focus on the role of learning strategies. The study found that mobile applications were more beneficial when students used them together with effective study strategies. This suggests that applications alone do not guarantee better academic performance. Instead, students need to use them in an organized and purposeful way.

Application design also affects how students develop productive study habits. Alrasheedi and Capretz [10] identified usability, collaboration, technical support, and instructional design as important factors in successful mobile learning. Applications that are easy to use, interactive, and well-supported are more likely to become part of students' regular

study routines. This is important because students may be less likely to continue using applications that are difficult to navigate, poorly organized, or not directly useful for their academic tasks.

Jacob and Issac [27] also examined the use of mobile devices for learning and found that mobile learning tools can support more consistent study habits when used with discipline and purpose. Their findings suggest that the value of mobile applications depends on how students include them in their study routines. For example, applications can help students review lessons, organize study materials, manage deadlines, and access learning resources more regularly. However, without self-regulation, the same devices may also become sources of distraction.

Caratiquit and Caratiquit [28] provided another perspective by examining ChatGPT as an academic support tool and its relationship with academic performance through learning motivation. Their study found that ChatGPT affected academic performance through students' learning motivation, which means that the tool itself was not the only factor involved. This supports the idea that study applications may contribute to academic outcomes when they encourage motivation, confidence, and active engagement. For the present study, this emphasizes the need to understand study application usage frequency together with students' learning behavior and academic purpose.

F. Philippine Studies on Digital Tools and Academic Performance

In the Philippine context, mobile learning, learning management systems, and educational applications have become increasingly relevant, especially after the shift to online and flexible learning. Tus et al. [18] examined LMS usage and academic performance among Filipino students during online learning. Their study found that LMS tools were used by students and were perceived as helpful in the online learning setup. However, LMS usage was not significantly related to academic performance. This finding is important because it suggests that the presence and use of digital learning platforms do not automatically result in higher academic performance.

Palmerola et al. [29] conducted a comparative study on learning modalities and academic performance in a private senior high school in Southern Philippines. Their findings showed that students' performance was influenced not only by the learning modality, but also by technological support, teacher guidance, and access to learning resources. This is relevant to the present study because students' use of study applications may also depend on their access to devices, internet connection, and familiarity with digital tools.

Sicangco et al. [30] examined the use of real-world applications in Technology and Livelihood Education instruction. Their study found that application-based learning tools helped improve students' performance and motivation.

Although the study focused on TLE, it supports the idea that digital tools can make learning more practical and engaging when they are connected to actual academic tasks.

Graciano and Benablo [31] focused on BSIT students by examining digital literacy skills and online information-searching strategies as determinants of academic performance. Their study found that both digital literacy and information-searching strategies significantly predicted academic achievement. This is highly relevant to the present study because BSIT students are expected to use digital tools for research, programming, collaboration, and other academic requirements. It also suggests that students with stronger digital skills may benefit more from study applications.

Pedraza and Canoy [32] assessed the effectiveness of mobile learning applications in selected secondary public schools in General Trias City, Philippines. Their findings showed that mobile learning applications supported student learning and engagement, particularly through interactive features, exam preparation, accessibility, and gamified elements. The study also found that socioeconomic status and usage frequency affected perceived effectiveness, suggesting that access and regular use can influence how much students benefit from mobile learning applications.

Santiago et al. [33] examined flexible learning adaptability among students at Cavite State University. Their study found that smartphones and mobile applications were among the most used educational tools. Google Classroom and Google Meet were also widely used for asynchronous and synchronous learning. The study emphasized that preparedness was an important factor in learning engagement. This reflects the experience of Filipino students and supports the importance of examining how students use digital applications for academic purposes.

Mamolo [34] developed and evaluated the Digital Interactive Math Comics (DIMaC) mobile application among Grade 11 students in the Philippines. Students reported positive learning experiences and considered the application acceptable for classroom use. Although the study focused on mathematics, it shows that locally developed mobile learning tools can support student engagement and make learning more interesting.

Barrientos [35] conducted a quasi-experimental study on the use of math applications and the mathematics performance of Grade 8 students in the Philippines. The study found that students' performance improved significantly after using math applications. This supports the idea that educational applications can positively affect academic performance when used as part of structured learning activities.

Overall, Philippine studies show that students use mobile applications, LMS platforms, and other digital tools to support academic work. However, these studies also show that the

benefits of study applications may depend on access, preparedness, digital literacy, learning motivation, socioeconomic conditions, and the purpose of use. These findings are relevant to BSIT students because their academic tasks often require the frequent and purposeful use of digital technologies.

G. Synthesis of Related Literature and Research Gap

The reviewed studies provide a clear basis for understanding the relationship between mobile learning, study applications, and academic performance. Previous review-based studies show that mobile learning generally has a positive effect on students' learning outcomes [6]– [8], [11], [12]. However, the literature also shows that its effectiveness depends on several factors, such as application design, usability, motivation, instructional support, learning strategies, and the purpose of use [10], [14], [21], [26], [28]. This means that mobile learning and study applications may support academic performance, but their impact may vary depending on how students use them and how well these tools support learning.

Studies on usage frequency also show mixed findings. Some studies found that application engagement or frequent use of mobile learning apps was related to academic performance or perceived learning benefits [19], [32]. However, other studies found that LMS usage or mobile application usage frequency was not significantly related to academic performance or perceived learning outcomes [18], [20]. These findings suggest that the academic value of study application usage may depend not only on how often students use the applications, but also on the type of application used, the purpose of use, and the quality of the learning experience.

The literature also shows that students use different categories of study applications for different academic needs. These include learning management systems, communication tools, collaboration platforms, content creation tools, assessment applications, productivity applications, reference tools, and subject-specific learning applications [22]– [25]. These categories are important because each type of application supports learning in a different way. For instance, productivity applications may help students manage tasks and deadlines, while subject-specific applications may help them practice skills, solve problems, or review course content.

In the Philippine context, local studies show that mobile applications, learning management systems, and other digital learning tools are commonly used to support students' academic activities [18], [29]– [35]. However, these studies also emphasize that the benefits of digital learning tools may be affected by access to technology, socioeconomic status, preparedness, digital literacy, learning motivation, and responsible use [28], [31]– [33]. Despite these findings, there is still limited research that directly examines the relationship between study application usage frequency and academic performance among Filipino BSIT students.

Most existing studies focus on mobile learning in general, smartphone use, LMS usage, or subject-specific applications. Only a limited number of studies treat study application usage frequency as the main variable, and fewer studies examine its relationship with academic performance measured through GWA in a BSIT context. Therefore, the present study addresses this gap by examining the relationship between study application usage frequency and the academic performance of second-year BSIT students at National University–Manila.

III. METHODOLOGY

This section presents the methods and procedures used in conducting the study. It discusses the research design, respondents of the study, research instrument, data collection procedure, and statistical treatment of data. These procedures were used to describe the study application usage patterns of second-year BSIT students at National University–Manila and determine the relationship between study application usage frequency and academic performance.

A. Research Design

This study used a quantitative research design, specifically a descriptive-correlational approach. The descriptive method was used to describe the respondents' study application usage patterns, including their frequency of use, categories of applications used, average daily usage, level of agreement on usage behavior and perceived learning impact, and academic performance in terms of self-reported General Weighted Average (GWA) range.

The correlational approach was applied to determine whether there was a significant relationship between study application usage frequency and the academic performance of the respondents.

B. Respondents of the Study

The respondents of this study were second-year Bachelor of Science in Information Technology students at National University–Manila. The respondents came from BSIT specializations, including Mobile and Web Applications and Multimedia Arts and Animation. They were considered suitable for the study because their academic tasks often involve the use of digital tools, online learning platforms, and study applications.

A total of 100 second-year BSIT students participated in the study. Convenience sampling was used because the respondents were selected based on their availability, accessibility, and willingness to participate within the given time frame. This sampling method was applied due to time and accessibility considerations. However, since the respondents were not selected through random sampling, the findings may not fully represent the entire population of second-year BSIT students at National University–Manila. Therefore, the results should be interpreted only within the context of the respondents who participated in the study and should not be generalized to all BSIT students or to the entire student population.

Only students who met the following criteria were included in the study:

1. Currently enrolled as a second-year BSIT student at National University–Manila; and
2. Willing to participate in the study.

C. Research Instrument

The study used a structured questionnaire as the primary instrument for data collection. The questionnaire was prepared based on the objectives of the study and was submitted to the subject professor for review and approval before the survey link was distributed to the respondents. This was done to ensure that the items were clear and aligned with the purpose of the study.

The questionnaire was divided into three sections.

1) Respondent Profile: This section gathered demographic and academic information from the respondents, such as their specialization and self-reported General Weighted Average (GWA) range.

2) Study Application Usage: This section collected information about the respondents' use of study applications. It included whether they used study applications for academic purposes, their frequency of use, average daily usage in hours, categories of study applications used, and specific applications commonly used.

3) Study Application Usage and Its Effects on Learning: This section used a 5-point Likert scale to measure the respondents' level of agreement with statements related to their usage behavior and the perceived impact of study applications on learning and academic performance.

The scale was interpreted as follows: 5 – Strongly Agree, 4 – Agree, 3 – Neutral, 2 – Disagree, and 1 – Strongly Disagree.

D. Data Collection Procedure

Data were collected using an online survey questionnaire distributed through Microsoft Forms. The researchers prepared and finalized the questionnaire based on the objectives of the study. The questionnaire included items about the respondents' profile, study application usage frequency, average daily usage, categories of study applications used, commonly used platforms, usage behavior, perceived learning impact, and self-reported General Weighted Average (GWA) range.

After the questionnaire was finalized, the survey link was shared with accessible second-year BSIT students at National University–Manila, specifically those from the Mobile and Web Applications (MWA) and Multimedia Arts and Animation (MAA) specializations. Since convenience sampling was used, the respondents were selected based on their availability, accessibility, and willingness to participate within the data collection period.

Before answering the questionnaire, the respondents were informed about the purpose of the study, the voluntary nature of their participation, and the confidentiality of their responses.

They were also informed that their responses would be used solely for academic purposes. After the target number of 100 responses was achieved, the form was closed. The collected data were then reviewed, coded, organized, and prepared for descriptive and correlational statistical analysis.

E. Statistical Treatment of Data

The collected data were analyzed using descriptive and inferential statistics. Frequency and percentage were used to describe the respondents' profile and study application usage patterns, including specialization, GWA range, use of study applications, frequency of use, average daily usage, categories of applications used, and commonly used study applications or platforms. For multiple-response questions, each selected category or application was counted separately; therefore, the total percentage could exceed 100%.

For the Likert-scale responses, answers were coded numerically, with Strongly Agree as 5, Agree as 4, Neutral as 3, Disagree as 2, and Strongly Disagree as 1. Mean, median, and mode were used to describe the respondents' level of agreement, while standard deviation, skewness, and kurtosis were used to examine the variability and distribution of the responses.

Pearson Product-Moment Correlation Coefficient was used to determine whether there was a significant relationship between study application usage frequency and academic performance. Study application usage frequency was coded from 1 to 5, with 1 representing Never and 5 representing Always. Academic performance was based on the respondents' self-reported GWA range, which was also coded from 1 to 5, with higher codes representing higher GWA ranges.

A significance level of 0.05 was used in testing the hypothesis. A p-value less than or equal to 0.05 indicated a statistically significant relationship, while a p-value greater than 0.05 indicated no statistically significant relationship. The computed r-value was used to interpret the strength and direction of the relationship.

IV. RESULTS

This section presents the descriptive statistics, statistical analysis results, and findings based on the responses of 100 second-year BSIT students at National University–Manila. All valid responses were included in the analysis.

A. Profile of the Respondents

TABLE I. DISTRIBUTION OF RESPONDENTS BY SPECIALIZATION

<i>Specialization</i>	<i>Frequency</i>	<i>Percentage</i>
BSIT – Mobile and Web Applications (MWA)	55	55%
BSIT – Multimedia Arts and Animation (MAA)	45	45%
Total	100	100.00%

Table I presents the distribution of respondents according to specialization. Out of 100 respondents, 55% were from BSIT–Mobile and Web Applications (MWA), while 45% were from

BSIT–Multimedia Arts and Animation (MAA). This indicates that both specializations were represented in the study, with a slightly higher number of respondents from the MWA specialization.

TABLE II. DISTRIBUTION OF RESPONDENTS BY GWA RANGE

<i>GWA Range</i>	<i>Frequency</i>	<i>Percentage</i>
3.50–4.00	43	43%
3.00–3.49	36	36%
2.50–2.99	12	12%
2.00–2.49	6	6%
1.50–1.99	3	3%
Total	100	100.00%

Table II presents the distribution of respondents according to their General Weighted Average (GWA) range. Out of 100 respondents, 43 were in the 3.50–4.00 range, while 36 were in the 3.00–3.49 range. Meanwhile, 12 respondents were in the 2.50–2.99 range, 6 were in the 2.00–2.49 range, and 3 were in the 1.50–1.99 range. This indicates that most respondents reported high academic performance based on the NU grading system, where a higher GWA represents better academic performance.

B. Study Application Usage

TABLE III. USE OF STUDY APPLICATIONS FOR ACADEMIC PURPOSES

<i>Response</i>	<i>Frequency</i>	<i>Percentage</i>
Yes	93	93%
No	7	7%
Total	100	100.00%

Table III presents the proportion of respondents who used study applications for academic purposes. Out of 100 respondents, 93% answered Yes, while 7% answered No. This indicates that most second-year BSIT respondents used study applications as part of their academic activities.

TABLE IV. FREQUENCY OF STUDY APPLICATION USAGE

<i>Frequency of Use</i>	<i>Frequency</i>	<i>Percentage</i>
Never	4	4%
Rarely	9	9%
Sometimes	30	30%
Often	34	34%
Always	23	23%
Total	100	100.00%

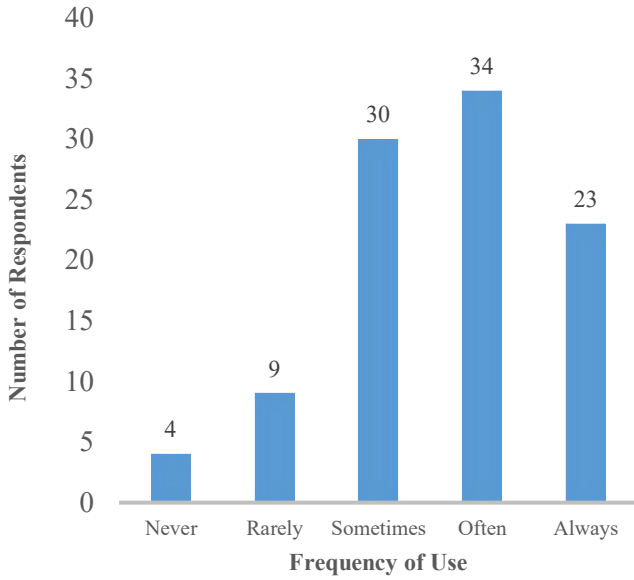


Fig. 1. Frequency of study application usage.

Fig. 1 illustrates how often the respondents used study applications for their studies. Out of 100 respondents, the highest number answered Often, with 34 respondents, followed by Sometimes, with 30 respondents. Meanwhile, 23 respondents answered Always, 9 respondents answered Rarely, and 4 respondents answered Never. This shows that most respondents used study applications at least sometimes for academic purposes, with a large portion using them often or always.

TABLE V. AVERAGE DAILY USAGE OF STUDY APPLICATIONS

<i>Average Daily Usage</i>	<i>Frequency</i>	<i>Percentage</i>
0 hours / Not applicable	5	5%
Less than 1 hour	15	15%
1–2 hours	31	31%
3–4 hours	26	26%
5–6 hours	18	18%
More than 6 hours	5	5%
Total	100	100.00%

Table V presents the average number of hours per day that respondents used study applications. Most respondents used study applications for 1–2 hours per day, with 31 respondents, followed by 3–4 hours per day, with 26 respondents. Meanwhile, 18 respondents used study applications for 5–6 hours per day, 15 used them for less than 1 hour, 5 used them for more than 6 hours, and another 5 answered 0 hours or not applicable. This indicates that many respondents used study applications for a moderate amount of time each day, particularly within the 1–4 hour range.

TABLE VI. CATEGORIES OF STUDY APPLICATIONS USED

<i>Category</i>	<i>Frequency</i>	<i>Percentage</i>
Learning platforms	71	71%
Collaboration tools	75	75%
Note-taking apps	51	51%
Coding practice tools	28	28%
Time management apps	22	22%
None / Not applicable	6	6%
Others	3	3%

Note: Multiple responses were allowed; therefore, the total percentage may exceed 100%.

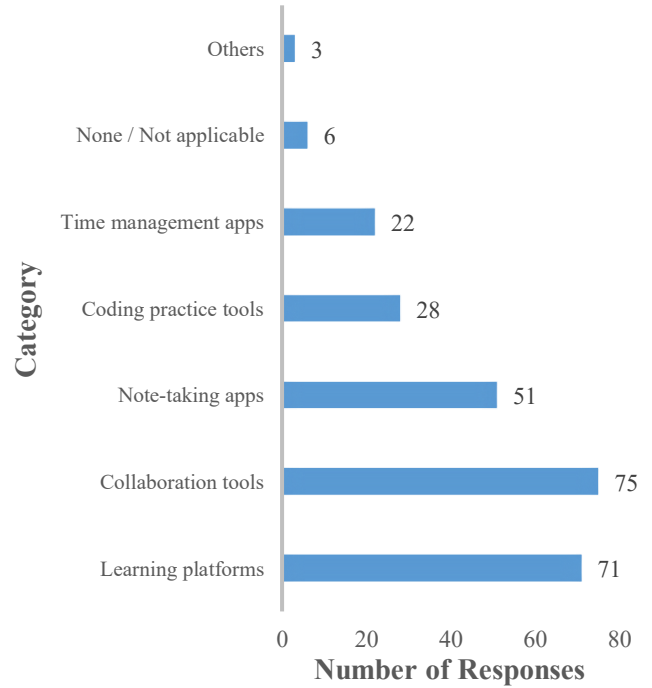


Fig. 2. Categories of study applications used by respondents.

Fig. 2 illustrates the categories of study applications used by the respondents. Since respondents were allowed to select more than one category, the total percentage exceeds 100%. The most commonly used category was collaboration tools, with 75 responses, followed by learning platforms with 71 responses, and note-taking apps with 51 responses. Meanwhile, 28 respondents selected coding practice tools, 22 selected time management apps, 6 answered none or not applicable, and 3 selected others. This suggests that respondents commonly used study applications for collaborating with their blockmates, accessing learning materials, and organizing academic notes.

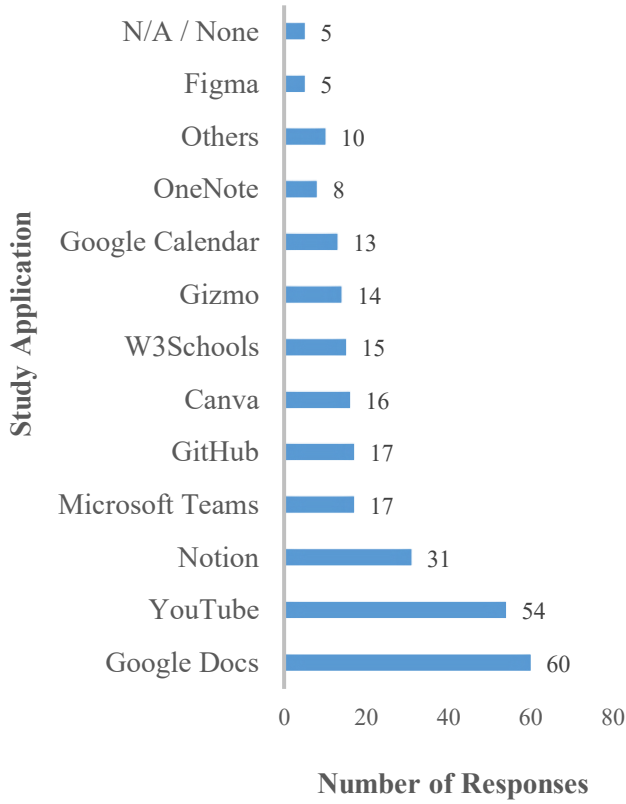


Fig. 3. Commonly used study applications among respondents.

Fig. 3 shows the commonly used study applications and platforms mentioned by the respondents. The results show that Google Docs was the most frequently mentioned platform, with 60 responses, followed by YouTube with 54 responses, and Notion with 31 responses. This indicates that many respondents commonly rely on tools that support collaboration, access to learning materials, and organization of notes. Other platforms such as Microsoft Teams, GitHub, Canva, W3Schools, Gizmo, Google Calendar, and OneNote were also mentioned, showing that respondents use different applications depending on their academic needs. Since respondents were allowed to mention more than one application or platform, the total percentage exceeds 100%.

C. Study Application Usage and Its Effects on Learning

The following items were used to measure the respondents' study application usage behavior:

UB1 – I use study applications when reviewing for quizzes or exams.

UB2 – I use study applications when completing assignments or projects.

UB3 – I use study applications to organize my notes and study materials.

UB4 – I use study applications to practice programming or technical skills.

UB5 – I use study applications consistently throughout the semester.

TABLE VII. CENTRAL TENDENCY OF STUDY APPLICATION USAGE BEHAVIOR

<i>Item</i>	<i>Mean</i>	<i>Median</i>	<i>Mode</i>
UB1	4.06	4.00	5.00
UB2	4.07	4.00	4.00
UB3	3.96	4.00	4.00
UB4	3.92	4.00	4.00
UB5	3.88	4.00	4.00

Table VII shows the central tendency of the respondents' study application usage behavior. Among the five items, UB2 obtained the highest mean of 4.07, indicating that respondents agreed that they use study applications when completing assignments or projects. This was closely followed by UB1, with a mean of 4.06, showing that respondents also agreed that they use study applications when reviewing for quizzes or exams. Meanwhile, UB5 obtained the lowest mean of 3.88, but it was still interpreted as Agree, indicating that respondents generally use study applications consistently throughout the semester.

The median for all items was 4.00, which means that the typical response of the respondents was Agree. In terms of mode, most items had a mode of 4.00, except UB1, which had a mode of 5.00. This shows that the most frequent response for using study applications when reviewing for quizzes or exams was Strongly Agree. Overall, the results suggest that respondents generally use study applications for academic-related activities, especially for completing assignments, reviewing for exams, organizing study materials, and supporting technical skill practice.

TABLE VIII. VARIABILITY AND DISTRIBUTION OF STUDY APPLICATION USAGE BEHAVIOR

<i>Item</i>	<i>Standard Deviation</i>	<i>Skewness</i>	<i>Kurtosis</i>
UB1	0.94	-0.86	0.29
UB2	0.93	-0.97	0.59
UB3	0.94	-0.88	1.05
UB4	1.02	-1.05	0.97
UB5	0.95	-0.78	0.50

Table VIII presents the variability and distribution of the respondents' study application usage behavior. The standard deviation values ranged from 0.93 to 1.02, indicating that the responses were moderately spread around the mean. Among the items, UB4 had the highest standard deviation of 1.02, showing that respondents had slightly more varied answers regarding the use of study applications for practicing programming or technical skills. Meanwhile, UB2 had the lowest standard deviation of 0.93, suggesting that responses for using study applications when completing assignments or projects were slightly more consistent.

The skewness values were all negative, ranging from -0.78 to -1.05, which means that most responses were concentrated on the higher end of the scale, particularly Agree and Strongly Agree. This supports the result that respondents generally showed positive usage behavior toward study applications. The kurtosis values ranged from 0.29 to 1.05, indicating that the response distributions were slightly to moderately peaked, with responses mostly clustered around higher agreement levels. Overall, the results suggest that respondents consistently agreed that they use study applications for various academic activities.

The following items were used to measure the respondents' perceived impact of study applications on learning:

- PI1 – Study applications help me manage my study time effectively.
- PI2 – Study applications improve my understanding of lessons.
- PI3 – Study applications help me complete academic tasks more efficiently.
- PI4 – Study applications increase my productivity in studying.
- PI5 – Study applications help me achieve better academic performance.

TABLE IX. CENTRAL TENDENCY OF THE PERCEIVED IMPACT OF STUDY APPLICATIONS

<i>Item</i>	<i>Mean</i>	<i>Median</i>	<i>Mode</i>
PI1	3.98	4.00	4.00
PI2	4.17	4.00	4.00
PI3	4.05	4.00	5.00
PI4	4.09	4.00	4.00
PI5	4.11	4.00	5.00

Table IX shows the central tendency of the respondents' perceived impact of study applications. Among the five items, PI2 obtained the highest mean of 4.17, indicating that respondents agreed that study applications improve their understanding of lessons. This was followed by PI5, with a mean of 4.11, showing that respondents also agreed that study applications help them achieve better academic performance. Meanwhile, PI1 obtained the lowest mean of 3.98, but it was still interpreted as Agree, indicating that respondents generally believed that study applications help them manage their study time effectively.

The median for all items was 4.00, which means that the typical response of the respondents was Agree. In terms of mode, PI3 and PI5 had a mode of 5.00, showing that the most frequent response for these items was Strongly Agree. Overall, the results suggest that respondents perceived study applications as helpful in improving lesson understanding, increasing productivity, completing academic tasks efficiently, and supporting academic performance.

TABLE X. VARIABILITY AND DISTRIBUTION OF THE PERCEIVED IMPACT OF STUDY APPLICATIONS

<i>Item</i>	<i>Standard Deviation</i>	<i>Skewness</i>	<i>Kurtosis</i>
PI1	0.80	-0.92	1.54
PI2	0.77	-0.85	1.52
PI3	0.95	-0.83	0.18
PI4	0.78	-0.68	0.31
PI5	0.94	-1.11	1.20

Table X presents the variability and distribution of the respondents' perceived impact of study applications. The standard deviation values ranged from 0.77 to 0.95, indicating that the responses were generally close to the mean but still showed slight variation among respondents. PI3 had the highest standard deviation of 0.95, suggesting that responses were slightly more varied regarding whether study applications help respondents complete academic tasks more efficiently. Meanwhile, PI2 had the lowest standard deviation of 0.77, showing that responses were more consistent regarding the improvement of lesson understanding.

The skewness values were all negative, ranging from -0.68 to -1.11, which means that most responses were concentrated on the higher end of the scale, particularly Agree and Strongly Agree. PI5 had the highest negative skewness of -1.11, indicating that many respondents agreed or strongly agreed that study applications help them achieve better academic performance. The kurtosis values ranged from 0.18 to 1.54, suggesting that the distributions were slightly to moderately peaked, with responses generally clustered around higher agreement levels. Overall, the results support the finding that respondents perceived study applications as beneficial to their learning, productivity, and academic performance.

D. Relationship Between Study Application Usage Frequency and Academic Performance

TABLE XI. PEARSON CORRELATION BETWEEN STUDY APPLICATION USAGE FREQUENCY AND ACADEMIC PERFORMANCE

<i>Variables</i>	<i>r-value</i>	<i>p-value</i>	<i>Decision</i>
Study Application Usage Frequency and GWA Range	0.58	<0.001	Reject H ₀

Pearson Product-Moment Correlation was used to determine the relationship between study application usage frequency and academic performance in terms of General Weighted Average (GWA) range. As shown in Table XI, the result obtained an r-value of 0.58, indicating a moderate positive correlation between the two variables. This means that as the frequency of study application usage increases, the respondents tend to belong to higher GWA ranges.

The computed p-value was less than 0.001, which is lower than the 0.05 level of significance. Therefore, the null

hypothesis was rejected. This indicates that there is a statistically significant relationship between study application usage frequency and academic performance among the respondents.

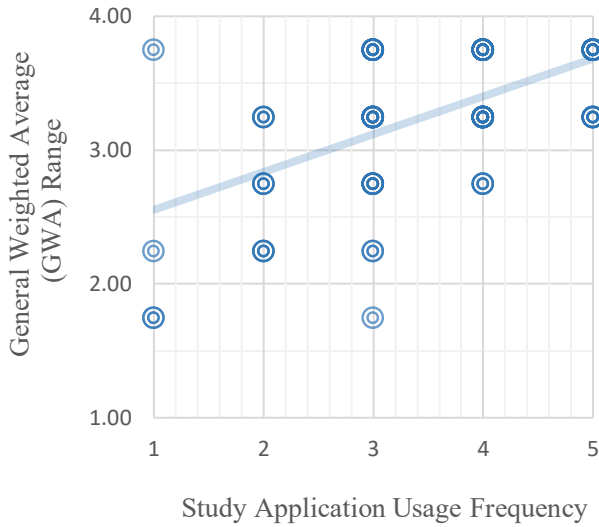


Fig. 4. Relationship between study application usage frequency and GWA.

Fig. 4 illustrates the scatter plot showing the relationship between study application usage frequency and General Weighted Average (GWA) Range. The figure shows an upward trend, indicating that respondents who reported using study applications more frequently also tended to belong to higher GWA ranges. This is consistent with the Pearson correlation result shown in Table XI, where the r -value of 0.58 indicates a moderate positive correlation between the two variables.

Although the data points are spread across the graph, the trendline suggests a clear positive direction. This means that higher study application usage frequency is associated with better academic performance among the respondents.

V. DISCUSSION

A. Interpretation of Results

The findings show that study applications are commonly used by second-year BSIT students at National University–Manila. Most respondents reported using study applications for academic purposes, with 93 out of 100 respondents answering “Yes.” In terms of frequency, most of the respondents used study applications sometimes, often, or always. This suggests that study applications have become part of the respondents’ academic routines.

The results also show that respondents used study applications for different academic needs. Collaboration tools, learning platforms, and note-taking applications were among the most selected categories. Google Docs, YouTube, and Notion were also the most frequently mentioned platforms. These findings suggest that respondents commonly use study applications for collaboration, accessing learning materials,

organizing notes, and completing academic tasks. This is related to the findings of Hinze et al. [22], who identified communication, storage, reference, and productivity tools as common categories of academic app use in higher education. It also supports Rangel-de Lazaro and Duarte [23], who emphasized that mobile learning applications may serve different academic functions, such as content creation, communication, collaboration, and assessment.

In terms of usage behavior, the respondents generally agreed that they use study applications for academic-related activities. The highest mean was obtained by the statement about using study applications when completing assignments or projects, followed closely by reviewing for quizzes or exams. This suggests that study applications are mostly used to support task completion and review activities.

For perceived impact, the respondents also generally agreed that study applications help them in their learning. The highest mean was obtained by the statement that study applications improve their understanding of lessons, followed by the statement that study applications help them achieve better academic performance. This finding is consistent with previous studies showing that mobile learning tools can support learning outcomes when they are useful, accessible, and aligned with students’ academic needs [11], [14]. Overall, the results suggest that respondents view study applications as helpful tools for improving lesson understanding, increasing productivity, completing academic tasks efficiently, and supporting academic performance.

B. Explanation of the Relationship Between Usage Frequency and Academic Performance

Pearson Product-Moment Correlation was used to determine the relationship between study application usage frequency and academic performance in terms of self-reported GWA range. The result showed an r -value of 0.58, indicating a moderate positive correlation between the two variables. This means that respondents who reported using study applications more frequently also tended to belong to higher GWA ranges.

The computed p -value was less than 0.001, which is lower than the 0.05 level of significance. Therefore, the null hypothesis was rejected. This indicates that there is a statistically significant relationship between study application usage frequency and academic performance among the respondents.

This finding suggests that study application usage frequency is related to academic performance. Students who use study applications more often may benefit from easier access to learning materials, better organization of notes, improved collaboration, and additional support for academic tasks. This result is consistent with Anwar et al. [19] and Pedraza and Canoy [32], who found that mobile learning

application engagement or usage frequency may be related to academic performance or perceived learning benefits.

However, the result should also be interpreted carefully because previous studies have shown mixed findings. For example, Tus et al. [18] found that LMS usage was not significantly related to academic performance, while Sharaf-Addin et al. [20] found that frequency of mobile application use did not create a statistically significant difference in perceived learning outcomes. These contrasting findings suggest that frequency alone may not fully explain academic performance. The type of application used, the purpose of use, learning motivation, app quality, and students' study habits may also influence the relationship between study application usage and academic outcomes.

Furthermore, the relationship found in this study should not be interpreted as direct causation. While frequent use of study applications is associated with a higher GWA range, academic performance may also be influenced by other factors such as motivation, time management, learning strategies, course difficulty, internet access, and consistency in using the applications effectively.

C. Implications, Limitations, and Future Recommendations

The findings imply that study applications can serve as helpful academic support tools for second-year BSIT students. Since the results showed a significant positive relationship between study application usage frequency and GWA range, students may benefit from using study applications more purposefully for reviewing lessons, organizing notes, managing tasks, collaborating with classmates, and practicing technical skills.

For students, the results suggest that study applications may support better learning outcomes when used consistently and effectively. For professors, the findings may provide insight into how students use digital tools in their academic work. This can help educators guide students in choosing appropriate applications for specific academic needs, such as collaboration, note-taking, programming practice, lesson review, and technical skill development.

However, the study has several limitations. Since convenience sampling was used, the respondents were selected based on availability and willingness to participate rather than through random selection. Because of this, the results may not fully represent the entire population of second-year BSIT students at National University–Manila. The findings are limited to the respondents who participated in the study and should not be generalized to all BSIT students, other year levels, or students from other programs and institutions.

In addition, the study relied on self-reported responses, including self-reported GWA range, which may be affected by recall or reporting bias. Since academic performance was collected by range rather than exact numerical GWA, the correlation result should be interpreted as an association

between coded categories rather than exact GWA scores. The study also focused only on the relationship between study application usage frequency and academic performance, while other possible factors such as motivation, study habits, time management, learning strategies, course difficulty, application quality, and internet access were not treated as main variables.

Future studies may use probability sampling methods, such as simple random sampling or stratified sampling, to improve the representativeness of the respondents. Researchers may also include a larger sample size, respondents from other year levels or programs, and exact GWA values instead of GWA ranges. Future researchers may also examine other factors such as study habits, motivation, learning strategies, application quality, type of application used, and purpose of application usage. This may provide a clearer understanding of how study applications contribute to students' learning and academic performance.

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